

Circular Economy Ontology Network (CEON) - Construction Module

Metadata

IRI	http://w3id.org/CEON/demo/construction/
Title	Circular Economy Ontology Network (CEON) - Construction Module
Publisher	Onto-DESIDE
Creator	Robin Keskisärkkä
Contributor	Eva Blomqvist Huanyu Li Mikael Lindecrantz
Date Created	2023-09-14
Date Issued	2025-06-30
License	https://creativecommons.org/licenses/by/4.0/
Version Iri	http://w3id.org/CEON/demo/construction/0.2/
Version Info	0.2
Prior Version	0.1
Preferred Namespace Prefix	ceon-demo-construction
Preferred Namespace Uri	http://w3id.org/CEON/demo/construction/
Description	The Product module of CEON (Circular Economy Ontology Network).

Classes

Condition ^C	
IRI	http://w3id.org/CEON/demo/construction/Condition
In Range Of	defineResourceCondition ^{op} hasCondition ^{op}
Super Class Of	FloorTileCondition ^C

Contaminant^C

IRI <http://w3id.org/CE0N/demo/construction/Contaminant>

In Range Of [hasContaminant^{OP}](#)

Dimensional Deviation^C

IRI <http://w3id.org/CE0N/demo/construction/DimensionalDeviation>

In Range Of [hasDimensionalDeviation^{OP}](#)

Fire Resistance Rating^C

IRI <http://w3id.org/CE0N/demo/construction/FireResistanceRating>

In Range Of [hasFireResistanceRating^{OP}](#)

Named Individuals
[E30ⁿⁱ](#)
[E60ⁿⁱ](#)

Fire Resistance Relation^C

IRI <http://w3id.org/CE0N/demo/construction/FireResistanceRelation>

Sub Class Of [actorODP:ResourceParticipation](#)

Floor Tile^C

IRI <http://w3id.org/CE0N/demo/construction/FloorTile>

Sub Class Of [product:Product](#)

Floor Tile Condition^C

IRI <http://w3id.org/CE0N/demo/construction/FloorTileCondition>

Sub Class Of [Condition^C](#)

Named Individuals
[bad_conditionⁿⁱ](#)
[good_conditionⁿⁱ](#)
[medium_conditionⁿⁱ](#)

Floor Tile Condition Relation^C

IRI <http://w3id.org/CE0N/demo/construction/FloorTileConditionRelation>

Sub Class Of [actorODP:ResourceParticipation](#)

Galvanized Steel Material^C

IRI <http://w3id.org/CE0N/demo/construction/GalvanizedSteelMaterial>

Sub Class Of [material:Material](#)

Policy On Treatment^C

IRI <http://w3id.org/CE0N/demo/construction/PolicyOnTreatment>

In Range Of [policyOnTreatment](#)^{op}

Named Individuals [refurbish](#)ⁿⁱ
[reprocess](#)ⁿⁱ
[reuse](#)ⁿⁱ

Product Quality^C

IRI <http://w3id.org/CE0N/demo/construction/ProductQuality>

In Range Of [defineResourceQuality](#)^{op}
[hasProductQuality](#)^{op}

Named Individuals [product_quality_high](#)ⁿⁱ
[product_quality_low](#)ⁿⁱ
[product_quality_medium](#)ⁿⁱ

Treatment Policy^C

IRI <http://w3id.org/CE0N/demo/construction/TreatmentPolicy>

Named Individuals [refurbish](#)ⁿⁱ
[reprocess](#)ⁿⁱ
[reuse](#)ⁿⁱ

Object Properties

associated process for lca^{op}

IRI <http://w3id.org/CE0N/demo/construction/associatedProcessForLCA>

define resource condition^{op}

IRI <http://w3id.org/CE0N/demo/construction/defineResourceCondition>

Domain [actorODP:Actor](#)

Range [Condition^c](#)

define resource quality^{op}

IRI <http://w3id.org/CE0N/demo/construction/defineResourceQuality>

Domain [actorODP:Actor](#)

Range [ProductQuality^c](#)

has condition^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasCondition>

Sub Property Of [topObjectProperty^{op}](#)

Domain [product:Product^c](#) or [product:Item^c](#)

Range [Condition^c](#)

has contaminant^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasContaminant>

Domain [product:Product](#)

Range [Contaminant^c](#)

has dimension^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasDimension>

Sub Property Of <http://qudt.org/schema/qudt/quantityValue>

Super Property Of

- [hasHeight^{op}](#)
- [hasThickness^{op}](#)
- [hasWidth^{op}](#)

Domain [product:Product^c](#) or [product:Item^c](#)

has dimensional deviation^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasDimensionalDeviation
Sub Property Of	topObjectProperty^{op}
Domain	product:Product
Range	DimensionalDeviation^c

has distance to recycling plant^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasDistanceToRecyclingPlant
Sub Property Of	http://qudt.org/schema/qudt/quantityValue

has fire resistance rating^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasFireResistanceRating
Sub Property Of	topObjectProperty^{op}
Domain	product:Product^c or product:Item^c
Range	FireResistanceRating^c

has height^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasHeight
Sub Property Of	hasDimension^{op}

has market value for new product^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasMarketValueForNewProduct
Sub Property Of	quantity:hasMarketValue^{op}
Domain	product:Product

has market value for reused product^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasMarketValueForReusedProduct
Sub Property Of	quantity:hasMarketValue^{op}
Domain	product:Product

has product quality^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasProductQuality
Sub Property Of	topObjectProperty^{op}
Domain	product:Item^c or product:Product^c
Range	ProductQuality^c

has recycling potential^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasRecyclingPotential
Sub Property Of	http://qudt.org/schema/qudt/quantityValue
Domain	product:Item^c or product:Product^c

has recycling separability^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasRecyclingSeparability
Sub Property Of	http://qudt.org/schema/qudt/quantityValue
Domain	product:Product^c or product:Item^c

has resistance to earth^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasResistanceToEarth
Sub Property Of	http://qudt.org/schema/qudt/quantityValue
Domain	product:Product

has reuse deconstructability^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasReuseDeconstructability
Sub Property Of	http://qudt.org/schema/qudt/quantityValue
Domain	product:Product

has system weight^{op}

IRI	http://w3id.org/CE0N/demo/construction/hasSystemWeight
Sub Property Of	http://qudt.org/schema/qudt/quantityValue
Domain	product:Product^c or product:Item^c

has thickness^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasThickness>

Sub Property Of [hasDimension^{op}](#)

has use span^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasUseSpan>

Sub Property Of <http://qudt.org/schema/qudt/quantityValue>

Super Property Of

- [hasUseSpanRemaining^{op}](#)
- [hasUseSpanTotal^{op}](#)

Domain [product:Item^c](#) or [product:Product^c](#)

has use span remaining^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasUseSpanRemaining>

Sub Property Of [hasUseSpan^{op}](#)

has use span total^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasUseSpanTotal>

Sub Property Of [hasUseSpan^{op}](#)

has width^{op}

IRI <http://w3id.org/CE0N/demo/construction/hasWidth>

Sub Property Of [hasDimension^{op}](#)

installation location^{op}

IRI <http://w3id.org/CE0N/demo/construction/installationLocation>

Sub Property Of [topObjectProperty^{op}](#)

policy on treatment^{op}

IRI <http://w3id.org/CE0N/demo/construction/policyOnTreatment>

Sub Property Of [topObjectProperty^{op}](#)

Domain [product:Product](#)

Range [PolicyOnTreatment^c](#)

has market value^{op}

IRI <http://w3id.org/CE0N/ontology/quantity#hasMarketValue>

Super Property Of

- [hasMarketValueForNewProduct](#)^{op}
- [hasMarketValueForReusedProduct](#)^{op}

Datatype Properties

construction year^{dp}

IRI <http://w3id.org/CE0N/demo/construction/constructionYear>

Sub Property Of [topDataProperty](#)^{dp}

Range [xsd:gYear](#)

has accessories^{dp}

IRI <http://w3id.org/CE0N/demo/construction/hasAccessories>

Domain [product:Product](#)^c or [product:Item](#)^c

Range [xsd:string](#)

has dismantling instruction^{dp}

IRI <http://w3id.org/CE0N/demo/construction/hasDismantlingInstruction>

Sub Property Of [topDataProperty](#)^{dp}

Range [xsd:anyURI](#)

has dismantling method^{dp}

IRI <http://w3id.org/CE0N/demo/construction/hasDismantlingMethod>

Sub Property Of [topDataProperty](#)^{dp}

Domain [product:Product](#)^c or [product:Item](#)^c

Range [xsd:string](#)

has installation date^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasInstallationDate
<u>Sub Property Of</u>	topDataProperty ^{dp}
<u>Domain</u>	product:Item
<u>Range</u>	xsd:date

has packaging recommendations after dismantling^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasPackagingRecommendationsAfterDismantling
<u>Sub Property Of</u>	topDataProperty ^{dp}
<u>Domain</u>	product:Product ^c or product:Item ^c
<u>Range</u>	xsd:string

has product description^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasProductDescription
<u>Sub Property Of</u>	topDataProperty ^{dp}
<u>Domain</u>	product:Product ^c or product:Item ^c
<u>Range</u>	xsd:string

has product id^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasProductID
<u>Sub Property Of</u>	topDataProperty ^{dp}
<u>Domain</u>	product:Item ^c or product:Product ^c
<u>Range</u>	xsd:int

has production date^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasProductionDate
<u>Sub Property Of</u>	topDataProperty ^{dp}
<u>Domain</u>	product:Item
<u>Range</u>	xsd:date

has production site^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasProductionSite
Sub Property Of	topDataProperty ^{dp}
Domain	product:Item ^c or product:Product ^c
Range	xsd:string

has refurbishment method^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasRefurbishmentMethod
Sub Property Of	topDataProperty ^{dp}
Domain	product:Item ^c or product:Product ^c
Range	xsd:string

has take back program from original manufacturer^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasTakeBackProgramFromOriginalManufacturer
Sub Property Of	topDataProperty ^{dp}
Domain	product:Product
Range	xsd:boolean

has take back program other manufacturer^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasTakeBackProgramOtherManufacturer
Sub Property Of	topDataProperty ^{dp}
Domain	product:Product
Range	xsd:boolean

has technical requirements for take back for recycling^{dp}

IRI	http://w3id.org/CE0N/demo/construction/hasTechnicalRequirementsForTakeBackForRecycling
Sub Property Of	topDataProperty ^{dp}
Domain	product:Item ^c or product:Product ^c
Range	xsd:string

has technical requirements for take back for reuse^{dp}

IRI <http://w3id.org/CE0N/demo/construction/hasTechnicalRequirementsForTakeBackForReuse>

Sub Property Of [topDataProperty](#)^{dp}

Domain [product:Product](#)^c or [product:Item](#)^c

Range [xsd:string](#)

is contaminated^{dp}

IRI <http://w3id.org/CE0N/demo/construction/isContaminated>

Sub Property Of [topDataProperty](#)^{dp}

Domain [product:Product](#)^c or [product:Item](#)^c

Range [xsd:boolean](#)

Annotation Properties

issued^{ap}

IRI <http://purl.org/dc/terms/issued>

publisher^{ap}

IRI <http://purl.org/dc/terms/publisher>

Namespaces

:

<http://w3id.org/CE0N/demo/construction/>

actor

<http://w3id.org/CE0N/ontology/actor/>

actorODP

<http://w3id.org/CE0N/ontology/actorODP/>

datasheet

<http://w3id.org/CE0N/ontology/datasheet/>

dcterms

<http://purl.org/dc/terms/>

material

<http://w3id.org/CE0N/ontology/material/>

owl

<http://www.w3.org/2002/07/owl#>

product

<http://w3id.org/CE0N/ontology/product/>

prov

<http://www.w3.org/ns/prov#>

quantity

<http://w3id.org/CEON/ontology/quantity#>

rdf

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

rdfs

<http://www.w3.org/2000/01/rdf-schema#>

vann

<http://purl.org/vocab/vann/>

xsd

<http://www.w3.org/2001/XMLSchema#>

Legend

c	Classes
op	Object Properties
dp	Datatype Properties
ap	Annotation Properties